Question-5.8.20

EE25BTECH11022 - sankeerthan

Question

Five years ago, Minu was thrice as old as Sonu.Ten years later,Minu will be twice as old as Sonu.How old are Minu and Sonu?

Solution

Let Minu's present age be x and Sonu's present age be y. Five years ago, Minu was thrice as old as Sonu:

$$\begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} - 5 = 3 \left(\begin{pmatrix} 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} - 5 \right) \tag{1}$$

$$\left(\begin{pmatrix} 1 & 0 \end{pmatrix} - 3 \begin{pmatrix} 0 & 1 \end{pmatrix} \right) \begin{pmatrix} x \\ y \end{pmatrix} = -10 \tag{2}$$

$$\begin{pmatrix} 1 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = -10 \tag{3}$$

Ten years later,Minu will be twice as old as Sonu:

$$\begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + 10 = 2 \left(\begin{pmatrix} 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + 10 \right) \tag{4}$$

$$\left(\begin{pmatrix} 1 & 0 \end{pmatrix} - 2 \begin{pmatrix} 0 & 1 \end{pmatrix} \right) \begin{pmatrix} x \\ y \end{pmatrix} = 10 \tag{5}$$

$$\begin{pmatrix} 1 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 10 \tag{6}$$

Matrix form:

$$\begin{pmatrix} 1 & -3 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -10 \\ 10 \end{pmatrix} \tag{7}$$

Let

$$\mathbf{A} = \begin{pmatrix} 1 & -3 \\ 1 & -2 \end{pmatrix}, \mathbf{X} = \begin{pmatrix} x \\ y \end{pmatrix}, \mathbf{B} = \begin{pmatrix} -10 \\ 10 \end{pmatrix} \tag{8}$$

Inverse of **A**:

$$\mathbf{A}^{-1} = \begin{pmatrix} -2 & 3 \\ -1 & 1 \end{pmatrix} \tag{9}$$

Solution

$$\mathbf{X} = \mathbf{A}^{-1}\mathbf{B} \tag{10}$$

$$= \begin{pmatrix} -2 & 3 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} -10 \\ 10 \end{pmatrix} \tag{11}$$

$$= \begin{pmatrix} 50 \\ 20 \end{pmatrix} \tag{12}$$

Therefore, Minu's present age $= 50$ and Sonu's present age $= 20$

C-Code

```c
// code.c
#include <stdio.h>

void solve_ages(double *minu, double *sonu) {
    double A[2][2] = {{1, -3}, {1, -2}};
    double B[2] = {-10, 10};
    double det = A[0][0]*A[1][1] - A[1][0]*A[0][1];

    // Inverse matrix elements
    double invA[2][2] = {{A[1][1]/det, -A[0][1]/det},
                         {-A[1][0]/det, A[0][0]/det}};

    // Multiply invA by B
    *minu = invA[0][0]*B[0] + invA[0][1]*B[1];
    *sonu = invA[1][0]*B[0] + invA[1][1]*B[1];
}
```

```python
# Code by GVV Sharma
# Modified for Problem Solution
# Released under GNU GPL
# Calculating area enclosed between curves
# call.py
import ctypes
import plot

lib = ctypes.CDLL('./code.so')
```

Python-Code

```python
minu = ctypes.c_double()
sonu = ctypes.c_double()

lib.solve_ages(ctypes.byref(minu), ctypes.byref(sonu))

print(f"Minu's age: {minu.value}")
print(f"Sonu's age: {sonu.value}")

# Pass Python values to plot script
plot.plot_ages(minu.value, sonu.value)
```